

5(a)	correct symbol for the heater or for the LDR	M1
	all correct symbols in series (ignore voltmeter) and no extra symbols	A1
	correct symbol for voltmeter and in parallel with the heater	B1
5(b)(i)	$R = \rho L / A$	C1
	$= (1.1 \times 10^{-6} \times 2.0) / 3.8 \times 10^{-7}$	A1
	$= 5.8 \Omega$	
5(b)(ii)	$I = 4.8 / 5.8$	A1
	$= 0.83 \text{ A}$	
5(c)(i)	A larger (for new wire) or $A \propto d^2$ (and d larger for new wire) or $R \propto 1 / d^2$ (and d larger for new wire)	M1
	so R is less (than that of first wire)	A1
5(c)(ii)	(heater / total resistance decreases so) current (in circuit) increases (so p.d. across LDR increases) or heater resistance decreases so it has a smaller share/proportion/fraction of the (total) voltage / e.m.f.	M1
	(so voltmeter) reading is less (than 4.8 V)	A1